



BANJUL CITY COUNCIL

SUNU Park

FUNCTIONAL REQUIREMENTS — FOR DEVELOPERS

Street parking for Banjul, on every phone.

What the SUNU Park prototype does, user by user: the driver and attendant apps, SMS and USSD, and the web portal for administrators, organisations, the police and the Council, and the technology it is built with.

Prototype release v1.2.50

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Built by INNOVII for Banjul City Council

Demo server

• TRY IT ON THE WEB

Home page	178.236.185.183
Administrator	/admin
Organisation	/org
Police	/police
Council	/council
SMS & USSD simulator	/sms

• INSTALL THE ANDROID APPS

SUNU Park (drivers)	APK
SUNU Park Officer (attendants)	APK
All releases & server package	GitHub
Source code	GitHub
Open APK links in Chrome; allow "Install unknown apps" once.	

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02 — OVERVIEW

Six kinds of users, five channels

Drivers pay for street parking from any phone; attendants check plates on their road; the Council sees the money. Drivers and attendants never use the web. Four kinds of staff and partners sign in to the portal.

USER	WHO THEY ARE	CHANNELS	SIGN-IN
Driver	Anyone parking in a marked SUNU Park bay; pays for any plate	Driver app · SMS 7275 · USSD *7275#	Phone number + 6-digit SMS code
Parking attendant	Council attendant on a fixed road and shift	Officer app · SMS · USSD from the registered number	Registered number + SMS code
Organisation	Bank, ministry or company covering its fleet	Web /org	Billing contact's phone + SMS code
Administrator	SUNU Park / Council staff running the service	Web /admin	Username + password
Police	Officers following up unpaid fines	Web /police (read-only)	Username + password
Council	Banjul City Council revenue office	Web /council (read-only)	Username + password

Business rules (the same in every channel)

DAILY PASS

200 GMD

Valid till 7pm the day it is paid. The pass follows the plate.

MONTHLY PASS

4,000 GMD

20 paid days; reminder SMS 3 days before it ends.

FINE

1,880 GMD

Added when a warning is not paid within 24 hours.

- **Paid hours** 7am–7pm, Monday–Saturday; free otherwise.
- **Warnings**: an unpaid car gets a warning; the driver pays the daily fee within 24 h, after that fee + fine.
- **Fines first**: a phone with an unpaid warning on any plate settles it before buying a new pass.
- **Offender history**: first-time or repeat offender, shown to attendants and police.

- **Organisations**: 48,000 GMD per car per year, paid upfront (less any discount); cars added later pay the months left.
- **Renewal** invoice 30 days before the year ends, 5 days' grace.
- **Revenue share**: 60% Council, 40% operator.
- Prices come from the **tariff**, never hard-coded; changes need a Council authority reference.

03 — DRIVERS

Pay the way you prefer

No cash and no account to open. Payment by Wave, Afrimoney, APS or QMoney (simulated on the demo server; switched off in production until providers are connected).

ID	DRIVER APP (ANDROID)	PRIORITY
DRV-01	Launch screen: navy brand screen that splits open into the app; Android splash matches its first frame.	Must
DRV-02	Onboarding & sign-in: three intro screens, phone number, 6-digit SMS code (expires in 5 min, 5 tries).	Must
DRV-03	Terms gate: read the terms to the end and accept before first use and after every new version.	Must
DRV-04	Home: status of each plate (PAID daily / monthly / organisation, or UNPAID), free-hours state, Council announcements.	Must
DRV-05	Pay: daily or monthly pass for any plate with a mobile-money provider; receipt with ticket number.	Must
DRV-06	Warnings: see open warnings, the amount owed and the deadline; pay them (offered first, other payments blocked).	Must
DRV-07	Plates: add and remove plates; see which are covered by an organisation.	Must
DRV-08	Activity & messages: payments, receipts and every SMS sent to the number.	Should
DRV-09	Profile: number, version, terms, support contacts, sign out.	Should



SMS line 7275

- Text a plate **BJL1234** → offer → reply **1** - **4** for the provider.
- M B JL1234** monthly pass · **HELP** · **TERMS** .
- Receipts, warning notices, reminders by SMS (under 160 characters where possible).



USSD *7275#

- Menu: 1 Daily pass · 2 Monthly pass · 3 Check a plate · 4 Pay a warning · 5 My plates · 6 Terms.
- Unpaid warning shown first. Works on any phone without internet.
- Gateway: **POST /api/ussd** (CON/END convention).



04 — PARKING ATTENDANTS

Check plates, not cash

ID	OFFICER APP (ANDROID) · ALSO BY SMS AND USSD FROM THE REGISTERED NUMBER	PRIORITY
ATT-01	Sign-in only for numbers registered in the back office; switched-off attendants are refused.	Must
ATT-02	Shift: start and end (START / END); the reply names the fixed road and shift (7am–1pm or 1pm–7pm).	Must
ATT-03	Check a plate: PAID (daily, monthly, organisation) or NOT PAID, with first-time / repeat-offender history.	Must
ATT-04	Issue a warning to an unpaid car (W B JL1234); the driver gets an SMS and an in-app notice.	Must
ATT-05	Checks & messages: today's checks, targets, messages from supervisors. Attendants never take money.	Should

04 — ORGANISATIONS · web/org

One invoice a year, every car covered

ID	ORGANISATION PORTAL	PRIORITY
ORG-01	Sign-in by the billing contact's phone and SMS code; no self sign-up (set up by the account manager).	Must
ORG-02	Overview: cars covered vs agreed, cover end date, open invoices, attendant checks this month.	Must
ORG-03	Fleet: add or remove cars one by one or upload a CSV list; added cars pay the months left.	Must
ORG-04	Invoices: yearly upfront per car, bank-transfer details, "I have paid" with reference; renewal 30 days before, 5 days' grace.	Must
ORG-05	Reports: statements of paid invoices (print / PDF) and checks per car.	Should

HOW THE USERS CONNECT · A WARNING, END TO END

1 · Attendant

Checks **BJL8080** : NOT PAID, first-time offender. Issues a warning.

2 · Driver

Gets an SMS and app notice: pay 200 GMD within 24 hours.

3 · After 24 h

Unpaid: 200 + 1,880 GMD. Reminder SMS; other passes blocked until paid.

4 · Police

Sees the overdue plate and its linked phone in Fines tracking.

5 · Council

Payment shows in the revenue report, 60% Council share.

Covered cars show PAID to attendants: drivers of an organisation's cars pay nothing. Organisations never see drivers' phone numbers or receipts.



Parking for Banjul, run from one place

ID	BACK OFFICE PAGE	REQUIREMENT
ADM-01	Dashboard	Live roads, attendants on shift, payments and checks today; items that need attention.
ADM-02	Attendants	Register (name, staff no., number, road, shift), move, switch off; welcome SMS.
ADM-03	Organisations	Create with billing contact, agreed cars and discount; see fleets and invoices.
ADM-04	Payments	Match bank transfers to invoices; record and refer payment exceptions.
ADM-05	Warnings	Record a warning paid at the Council office (receipt no.); cancel one issued by mistake.
ADM-06	Tariffs & rules	Daily price, yearly price per car, fine; needs a Council authority reference; history kept.
ADM-07	Announcements	Publish, show or hide notices in the driver app.
ADM-08	Terms & conditions	Edit and publish versions; every app asks to accept again; optional SMS with the link.
ADM-09	Staff & audit	Create Administrator, Police or Council accounts (temporary password shown once), reset, switch off; tamper-proof audit log.



Police · /police

read-only

- Fines tracking: open warnings, overdue fines (24 h passed) highlighted first.
- Plate record: first-time or repeat offender, every warning and its status.
- The phone numbers linked to a plate, for follow-up. No receipts or wallets.
- Cannot change anything; every sign-in recorded.



Council · /council

read-only

- Revenue report: collected today, Council share 60%, operator share 40%.
- By source (daily, monthly, organisations, warnings) and by road, with unpaid rate.
- Fines tracking and the terms in force; print or save as PDF.
- Cannot change anything; every sign-in recorded.

06 — FOR EVERY USER

Shared requirements



Sign-in & security

- Phone users: 6-digit SMS code, 5 minutes, 5 tries; one code per minute.
- Staff: first password is temporary and must be changed (10+ characters).
- 5 wrong passwords → account waits 15 minutes. Sessions last 12 hours.
- Every action checked on the server against the user's role and own number.



Privacy

- Each role receives only its own data (server-side views per role).
- Drivers: their plates and receipts. Attendants: their shift and checks.
- Organisations: their fleet. Police: warnings and linked numbers only.
- Nobody but staff sees other people's phone numbers.



Terms & conditions

- Versioned by administrators; public at </terms> (sunupark.gm/terms).
- Apps: must scroll to the end and tick before continuing.
- SMS/USSD: paying means accepting; **TERMS** sends the link.



Audit & live updates

- Every sign-in and change written to the audit log with the person's name.
- A check, payment or announcement reaches every open screen within seconds.
- Release number shown in apps, portal and </api/health>.



Notifications

- Receipt after every payment; warning SMS with deadline.
- Warning reminder the next day; monthly pass reminder 3 days before.
- Organisation renewal invoice; new terms by SMS (optional).



Demo & testing tools

- SMS & USSD simulator at </sms>: any phone, 18 ready use cases.
- Demo clock (move time, next day) and reset, demo servers only.
- Production mode starts empty, real time, payments off.

Look & language. Pale blue [#F6FBFE](#), navy [#0B2E63](#), SUNU yellow [#FEDB46](#), Plus Jakarta Sans; the Banjul scene as vector art. Plain words, prices always from the tariff. Web home page is landscape on wide screens and stacks on phones.

07 — HOW THE PROTOTYPE IS BUILT

Technology



LAYER	TECHNOLOGY	WHY
Apps	HTML/CSS/vanilla JavaScript wrapped with Capacitor into Android APK/AAB	One small codebase, no framework, fast on low-end phones
Web	Static HTML/CSS/JS pages served by Nginx ; vector SVG art; Plus Jakarta Sans	Sharp on every screen, nothing to install
Rules	<code>engine.js</code> and <code>ParkingEngine.java</code> , kept identical by parity tests	Same answers on server, apps and SMS
Server	Java 17 servlets on Tomcat 10.1 , Maven, Jackson, SLF4J/Log4j2	INNOVII standard stack, Rocky Linux
Data	MariaDB , versioned SQL migrations applied on start	Simple upgrades, nightly backups
Messaging	Kannel (SMPP) for SMS; Africa's Talking-style USSD endpoint; simulated mode for demos	Works with Gambian telcos
Payments	Wave, Afrimoney, APS, QMoney: simulated now ; organisations by bank transfer	Real APIs once agreements are in place
Quality	GitHub Actions : back-end tests with MariaDB, Playwright browser tests, CodeQL , Dependabot	Every merge is tested and released automatically
Releases	1.2.<build> : server package + both APKs on GitHub;	Repeatable installs



Demo sign-ins

- Police: `police` / `police-demo`
- Council: `council` / `council-demo`
- Organisation: Demo Bank, phone **7101234** (code on screen)
- Administrator: account from INNOVII



Next steps to go live

- Mobile-money provider APIs and real payments.
- Telco SMS short code and USSD gateway.
- HTTPS domain (sunupark.gm), Play Store upload key.
- Production server, backups, staff training.

Code & docs: github.com/abdiMuhamud/parkna-demo · README, docs/ARCHITECTURE.md, docs/DEPLOYMENT.md, CLAUDE.md.